

Michael DiPasquale

CONTACT INFORMATION	New Mexico State University Department of Mathematical Sciences	<i>Mobile:</i> 217-552-7673 <i>E-mail:</i> midipasq@nmsu.edu <i>WWW:</i> http://midipasq.github.io
RESEARCH INTERESTS	Computational commutative algebra and algebraic geometry. Emphasis on pure and applied problems which can be approached with the tools of algebraic geometry and commutative algebra.	
EDUCATION	University of Illinois Urbana-Champaign (UIUC), Urbana, IL Ph.D. , Mathematics, May 2015 <i>Advisor:</i> Professor Hal Schenck <i>Thesis:</i> Splines on polytopal complexes Wheaton College , Wheaton, IL B.S. , Mathematics, May 2009	
ACADEMIC APPOINTMENTS	New Mexico State University (NMSU), Las Cruces, NM <i>Assistant Professor</i> University of South Alabama (USA), Mobile, AL <i>Assistant Professor</i> Colorado State University (CSU), Fort Collins, CO <i>Postdoctoral Fellow</i> Oklahoma State University (OSU), Stillwater, OK <i>Visiting Assistant Professor</i>	August 2023 - August 2021 - May 2023 August 2018 - July 2021 August 2015 - May 2018
PUBLICATIONS	27. <i>Geometric aspects of the Jacobian of a hyperplane arrangement</i> (with J. Sidman and W. Traves), Int. Math. Res. Not. IMRN (2025), no. 13, rnaf172.. arXiv:2209.04929. 26. <i>Restriction and extension for planar splines on triangulations</i>, in Lanini, Manni, and Schenck, editors, <i>Approximation Theory and Numerical Analysis meet Algebra, Geometry, Topology</i>, Springer INdAM Series 60 (2024), 149-172. Springer Nature Link page 25. <i>Planar splines on a triangulation with a single totally interior edge</i> (with B. Yuan), SIAM J. Appl. Algebra Geom. 8 (2024), no. 3, 686–712. arXiv:2306.16825 24. <i>A lower bound for the dimension of tetrahedral splines in large degree</i> (with N. Villamizar), Constr. Approx. 59 (2024), no. 1, 1–30. arXiv:2007.12274 23. <i>Duality for asymptotic invariants of graded families</i> (with T. Nguyen and A. Secoleanu), Adv. Math. 430 (2023), Paper No. 109208, 47 pp. arXiv:2208.11110 22. <i>Quasi-polynomial growth of numerical and affine semigroups with constrained gaps</i> (with B. Gillespie and C. Peterson), Semigroup Forum 107 (2023), no. 1, 60–78. arXiv:2208.09760 21. <i>A homological characterization for freeness of multi-arrangements</i>, Math. Ann. 385 (2023), no. 1-2, 745–786. arXiv:1806.05295 20. <i>On resurgence via asymptotic resurgence</i> (with B. Drabkin), J. Algebra. 587 (2021), 64-84. arXiv:2003.06980 19. <i>Koszul multi-Rees algebras of principal L-Borel Ideals</i> (with B. Jabbar Nezhad), J. Algebra. 581 (2021), 353-385. arXiv:2008.09565 18. <i>A lower bound for splines on tetrahedral vertex stars</i> (with N. Villamizar), SIAM J. Appl. Algebra Geom. 5 (2021), no. 2, 250-277. arXiv:2005.13043 17. <i>Counting the dimension of splines of mixed smoothness: A general recipe, and its application to meshes of arbitrary topologies.</i> (with D. Toshniwal), Adv. Comput. Math. (2021) arXiv:2001.01774	

16. *On the apolar algebra of a product of linear forms* (with Z. Flores and C. Peterson). In *Proceedings of the 45th International Symposium on Symbolic and Algebraic Computation*, IS-SAC '20, pages 130-137, New York, NY, USA, 2020. Association for Computing Machinery, [arXiv:2002.04818](#)
15. *A Generalization of Wilf's Conjecture for Generalized Numerical Semigroups* (with C. Cisto, G. Failla, Z. Flores, C. Peterson, and R. Utano), *Semigroup Forum* 101 (2020). [arXiv:1909.13120](#)
14. *Bivariate Semialgebraic Splines* (with F. Sottile), *J. Approx. Theory* 254 (2020), 105392, 19 pp. [arXiv:1905.08438](#)
13. *Free and non-free multiplicities on the A_3 arrangement* (with C. Francisco, J. Mermin, and J. Schweig), *J. Algebra* 544 (2020), 498-532. [arXiv:1609.00337](#)
12. *Asymptotic resurgence via integral closures* (with C. Francisco, J. Mermin, and J. Schweig), *Trans. Amer. Math. Soc.* 372 (2019), no. 9, 6655-6676. [arXiv:1808.01547](#)
11. *The Rees algebra of a two-Borel ideal is Koszul* (with C. Francisco, J. Mermin, J. Schweig, and G. Sosa), *Proc. Amer. Math. Soc.* 147 (2019), no. 2, 467-479. [arXiv:1706.07462](#)
10. *Free multiplicities on the moduli of X_3* (with M. Wakefield), *J. Pure Appl. Algebra* 222 (2018), no. 11, 3345-3359. [arXiv:1707.03961](#)
9. *Inequalities for free multi-braid arrangements*, *Proc. Japan Acad. Ser. A Math. Sci.* 94 (2018), no. 4, 36-41. [arXiv:1705.02409](#)
8. *Dimension of mixed splines on polytopal cells*, *Math. Comp.* 87 (2018), no. 310, 905-939. [arXiv:1411.2176](#)
7. *Semialgebraic splines* (with F. Sottile and L. Sun), *Comput. Aided Geom. Design* 55 (2017), 26-47. [arXiv:1604.05947](#)
6. *Generalized splines and graphic arrangements*, *J. Algebraic Combin.* (2016), 1-19. [arXiv:1606.03091](#)
5. *Associated primes of spline complexes*, *J. Symb. Comput.* (2016), 158-199. [arXiv:1410.6894](#)
4. *Lattice-supported splines on polytopal complexes*, *Adv. in Appl. Math.* 55 (2014), 1-21. [arXiv:1312.3294](#)
3. *Shellability and freeness of continuous splines*, *J. Pure Appl. Algebra* 216 (2012), 2519-2523.
2. *Asymptotic connectivity of hyperbolic planar graphs* (with P. Bahls), *Discrete Math.* 310 (2010), 3462-3472.
1. *On the order of a group containing nontrivial Gassmann equivalent subgroups*, *Rose-Hulman Undergraduate Mathematics Journal* 10, Issue 1 (2009).
0. *Splines on polytopal complexes*. Thesis (Ph.D.) University of Illinois at Urbana-Champaign (2015). 148 pp. ISBN: 978-1339-32551-4, ProQuest LLC.

UNDER REVIEW

6. *Asymptotic Resurgence of Facet ideals of Graphic Matroids* (with L. Fouli and A. Kumar), submitted. [arXiv:2607.26294](#)
5. *Trivariate Splines on Fans of Hyperplane Arrangements and Koszul Homology* (with C. Checa, P. Mazón, T. T. Nguyễn, L. Sega, P. Udanshive, A. Van Tuyt, and N. Villamizar), submitted. [arXiv:2606.18298](#)
4. *Algebraic geometry, Bernstein-Bézier analysis, and multivariate spline spaces* (with P. Alfeld, M. Sirvent, T. Sorokina, N. Villamizar, W. Whiteley, and B. Yuan), submitted.
3. *Asymptotic Resurgence of facet and Stanley-Reisner ideals of matroids* (with L. Fouli and A. Kumar), submitted. [arXiv:2607.20892](#)
2. *Projective curves passing through finite sets and rank decompositions of symmetric tensors* (with C. Bocci, L. Chiantini, A. Mazzon, and C. Peterson), submitted.
1. *Generalized Hamming weights and symbolic powers of Stanley-Reisner ideals of matroids* (with L. Fouli, A. Kumar, and S. Tohăneanu), submitted. [arXiv:2406.13658](#).

EXTERNAL GRANTS

PI, NSF standard grant DMS-2201084/2344588 (2022-2026)
 AMS-Simons travel grant (2015-2018)

INTERNAL AWARDS	USA Support and Development Award (2022) \$1500 for bringing collaborators and speakers to USA USA Faculty Development Council Fellow (2022) \$5000 for research collaboration and development of external grant application																													
TEACHING EXPERIENCE	Instructor of record <table><tr><th>Course</th><th>Description</th></tr><tr><td>Abstract algebra II (NMSU)</td><td>second grad course in algebra</td></tr><tr><td>Commutative algebra and algebraic geometry (NMSU)</td><td>grad course on solving polynomial systems</td></tr><tr><td>Intro Modern Algebra (CSU, USA, NMSU)</td><td>group theory and proof writing</td></tr><tr><td>Intro to Higher Math (CSU, NMSU)</td><td>proof writing</td></tr><tr><td>Linear Algebra (CSU, USA, NMSU)</td><td>matrix theory</td></tr><tr><td>Precalculus Trigonometry (USA)</td><td>trigonometric functions and modeling</td></tr><tr><td>Finite Mathematics (USA)</td><td>probabilities, counting, and logic for non-math majors</td></tr><tr><td>Intro to Combinatorial Theory (CSU)</td><td>combinatorics and number theory</td></tr><tr><td>Calculus 2 (NMSU, CSU, USA)</td><td>sequences, series, and integration techniques</td></tr><tr><td>Intro to Real Analysis (OSU)</td><td>proof writing and real analysis</td></tr><tr><td>Calculus 1 (OSU)</td><td>differential and integral calculus</td></tr><tr><td>A Mathematical World (UIUC)</td><td>survey course emphasizing applications of mathematics</td></tr><tr><td>College Algebra (UIUC)</td><td>calculus preparation course</td></tr></table> - Responsible for lecturing, grading exams and quizzes, writing worksheets and homework - Wrote exams for most courses - Often implemented group work at least once per week		Course	Description	Abstract algebra II (NMSU)	second grad course in algebra	Commutative algebra and algebraic geometry (NMSU)	grad course on solving polynomial systems	Intro Modern Algebra (CSU, USA, NMSU)	group theory and proof writing	Intro to Higher Math (CSU, NMSU)	proof writing	Linear Algebra (CSU, USA, NMSU)	matrix theory	Precalculus Trigonometry (USA)	trigonometric functions and modeling	Finite Mathematics (USA)	probabilities, counting, and logic for non-math majors	Intro to Combinatorial Theory (CSU)	combinatorics and number theory	Calculus 2 (NMSU, CSU, USA)	sequences, series, and integration techniques	Intro to Real Analysis (OSU)	proof writing and real analysis	Calculus 1 (OSU)	differential and integral calculus	A Mathematical World (UIUC)	survey course emphasizing applications of mathematics	College Algebra (UIUC)	calculus preparation course
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	Recitation instructor , University of Illinois Urbana-Champaign - Led bi-weekly 50-minute problem sessions and proctored and graded quizzes and exams for seven semesters of Calculus (1,2, and 3) - Led student groups through worksheets I had written during bi-weekly two-hour workshops for one semester of Calculus 1 in the Merit program - Appeared on the ‘List of Teachers Ranked as Excellent’ by their students in three semesters																													
	Undergraduate teaching assistant , Wheaton College - Led problem sessions once per week at Wheaton College for Analysis I, Algebra I, and Discrete Mathematics																													
STUDENTS SUPERVISED	Ph.D. Advisor for Sarah Schott, James Warner (2025-) Research Assistantship for David Salomon (2025-2026) Research Assistantship for Jackson West (2024-2025) Non-thesis masters advisor for Sean Palmer (2023-2024) Masters thesis advisor for Ryann Firestine (2022-2023)																													
MENTORING	Co-mentor (with Nelly Villamizar) for focus group on <i>Approximation Theory</i> May 2025 during the Workshop on the Applications of Commutative Algebra at the Fields Institute, Toronto, Canada. This resulted in the paper <i>Trivariate Splines on Fans of Hyperplane Arrangements and Koszul Homology</i> . Assistant for a minicourse on Algebraic Geometry at SMI in Perugia Summer 2019 <i>Created problem sets and ran Macaulay2 help sessions twice per week.</i> Honors option for Intro to Math Reasoning, Linear Algebra, Calculus 2 Fall 2019, 2020, 2022 <i>Created additional problem sets and problem sessions for students to receive honors credit.</i>																													

Mentor in the [Illinois Geometry Lab](#)

Spring 2014, Fall 2014

*Co-led undergraduate research on minimal energy configurations of particles.***Teaching mentor** for junior graduate students

Fall 2013

*Mentored several first-year graduate students, visited classes and offered teaching feedback.*DISSEMINATION
OF RESEARCH

Lead co-author of the package [AlgebraicSplines](#) for the computer algebra system Macaulay2. This package is currently used by several researchers, including Julianna Tymoczko, who employs this package in research with undergraduates at Smith College.

CONFERENCE
PRESENTATIONS

1. Some results on the containment problem for Stanley-Reisner ideals of matroids 03/2026
Spring 2026 Western AMS Sectional Meeting
Special Session on Commutative Algebra, Algebraic Geometry and Applications
2. Resurgences for some Stanley-Reisner ideals of matroids 10/2025
Fall 2025 Eastern Virtual AMS Sectional Meeting
Special Session on Commutative Algebra and its Mathematical Interactions
3. Algebraic properties of k -generated arrangements 10/2025
AMS Sectional Meeting, Tulane, LA
Special Session on Hyperplane Arrangements and Polytopes
4. Splines and hyperplane arrangements 05/2025
Workshop on Applications of Commutative Algebra at the Fields Institute, Toronto, Canada
5. Generalized Hamming weights and symbolic powers of Stanley-Reisner ideals of matroids 02/2025
Workshop on Computational Interactions between Algebra, Combinatorics, and Discrete Geometry, Institute for Pure and Applied Mathematics (IPAM), CA
6. Rank equations for unexpected dimension of planar C^1 cubic splines 09/2024
AMS Sectional Meeting, San Antonio, TX
Special Session on Applications of Algebraic Geometry
7. A case study in bivariate semialgebraic splines 04/2024
AMS Sectional Meeting, Milwaukee, WI
Special Session on Connections between Commutative Algebra and Algebraic Combinatorics
8. Saturation of the Jacobian ideal of a hyperplane arrangement in minimal degree 12/2023
Hyperplane Arrangements 2023 at Rikkyo University, Tokyo, Japan
9. Lex-segment initial ideals and the dimension of planar splines 11/2023
SIAM Texas-Louisiana Sectional Meeting, Lafayette, Louisiana
Minisymposium on applications of combinatorial and computational algebraic geometry
10. Restriction and Extension for Planar Splines 07/2023
SIAM Conference on Applied Algebraic Geometry, Eindhoven, Netherlands
Minisymposium on Algebraic Spline Geometry
11. Dimension of bivariate splines on a partition with one totally interior edge 05/2023
International Conference on Approximation Theory and Beyond, Nashville, TN
Minisymposium on Multivariate Splines: Theory and applications
12. Apolarity for differentially closed filtrations of ideals 03/2023
AMS Sectional Meeting, Atlanta, GA
Special Session on Recent Developments in Commutative Algebra
13. Curves passing through space points and Waring rank 01/2023
Joint Mathematics Meetings, Boston, MA
AMS Special Session on Applied Enumerative Geometry
14. Singularities of line arrangements and rigidity of planar frameworks 12/2022
Virtual workshop organized by Mustapha Lahyane (University of Michoacán, Mexico)
Commutative Algebra, Algebraic Geometry and Related Topics
15. Homogeneous trivariate splines on the star of a vertex 09/2022
INdAM Meeting in Cortona, Italy
Approximation Theory and Numerical Analysis meet Algebra, Geometry, Topology

16. *Duality for sequences associated to symbolic powers* 05/2022
AMS Sectional Meeting, Denver, CO (virtual due to COVID-19)
Special Session on Commutative Algebra
17. *Saturating the Jacobian ideal of a line arrangement and parallel drawings* 3/2022
AMS Sectional Meeting, Purdue, IN (virtual due to COVID-19)
Special Session on Combinatorial Techniques in Commutative Algebra
18. *Rigidity, formality, and syzygies of the module of derivations of a line arrangement* 10/2021
AMS Sectional Meeting, Albuquerque, NM (virtual due to COVID-19)
Special Session on Hyperplane arrangements in connection with commutative algebra
19. *Curves passing through points in projective space* 10/2021
AMS Sectional Meeting, Omaha, NE (virtual due to COVID-19)
Special Session on Commutative Algebra
20. *Continuous splines on cross-cut cells and rigid planar frameworks* 08/2021
SIAM Conference on Applied Algebraic Geometry (virtual due to COVID-19)
Minisymposium on Algebraic Methods for Multivariate Splines and Rigidity
21. *Koszul multi-Rees algebras arising from principal Borel ideals* 03/2021
AMS Sectional Meeting, Providence, RI (virtual due to COVID-19)
Special Session on Current Trends in Combinatorial Commutative Algebra
22. *Dual sequences arising from apolarity* 03/2021
AMS Sectional Meeting, Atlanta, GA (virtual due to COVID-19)
Special Session on Commutative Algebra and its Interaction with Algebraic Geometry and Combinatorics
23. *Formal line arrangements and rigid planar frameworks* 01/2021
Mathematisches Forschungsinstitut Oberwolfach, Germany (virtual due to COVID-19)
Workshop on Logarithmic Vector Fields and Freeness of Divisors and Arrangements
24. *Regularity of uniform power ideals and the Waldschmidt constant* 10/2020
AMS Sectional Meeting, University Park, PA (virtual due to COVID-19)
Special Session on Commutative Algebra and Connections to Algebraic Geometry and Combinatorics
25. *On the apolar algebra of a product of linear forms* 07/2020
The 45th International Symposium on Symbolic and Algebraic Computation, ISSAC '20
(virtual due to COVID-19)
26. **(Cancelled due to COVID-19)** *Generalizing Wilf's conjecture to higher dimensions* 05/2020
AMS Sectional Meeting, Fresno, CA
Special Session on Numerical Semigroups and Applications
27. **(Cancelled due to COVID-19)** *A linear bound on the regularity of power ideals* 04/2020
AMS Sectional Meeting, West Lafayette, IN
Special Session on Combinatorial Techniques in Commutative Algebra
28. *A generalization of Wilf's Conjecture* 01/2020
AMS-MAA Joint Mathematics Meetings, Denver, CO
AMS Special Session on Recent Trends in Semigroup Theory
29. *Apolarity and trivariate piecewise polynomials* 08/2019
Algebraic Spline Geometry Meeting, Swansea, United Kingdom
30. *Algebraic Approaches to Spline Theory* 07/2019
SIAM Conference on Applied Algebraic Geometry, Bern, Switzerland
Minisymposium on Multivariate Spline Approximation and Algebraic Geometry
31. *Asymptotic Resurgence via Integral Closure and Linear Programs* 02/2019
Southwest Local Algebra Meeting, El Paso, TX
32. *Asymptotic Resurgence and Integral Closures* 11/2018
AMS Sectional Meeting, Fayetteville, AR
Special Session on Interactions Between Combinatorics and Commutative Algebra
33. *Freeness of Multi-arrangements via Acyclicity* 06/2018
Research Institute for Mathematical Sciences (RIMS), Kyoto, Japan
Matroids, reflection groups, and free hyperplane arrangements

34. *A Homological Approach to Freeness of Multi-arrangements* 04/2018
AMS Sectional Meeting, Boston, MA
Special Session on Arrangements of Hypersurfaces
35. *The Toric Ring of a Two-Borel ideal is Koszul* 01/2018
AMS-MAA Joint Mathematics Meetings, San Diego, CA
AMS Special Session on Combinatorial Commutative Algebra and Polytopes
36. *Freeness of Multi-Coxeter Arrangements of type A* 09/2017
AMS Sectional Meeting, Denton, TX
Special Session on Algebraic Combinatorics of Flag Varieties
37. *Splines on planar semi-algebraic partitions* 09/2017
AMS Sectional Meeting, Denton, TX
Special Session on Applicable and Computational Algebraic Geometry
38. *Algebraic Methods in Spline Theory* 08/2017
SIAM Conference on Applied Algebraic Geometry, Atlanta, GA
Minisymposium on Multivariate Splines and Algebraic Geometry
39. *Multi-derivations on the moduli of the X_3 arrangement* 04/2017
AMS Sectional Meeting, Pullman, WA
Special Session on Combinatorial and Computational Commutative Algebra and Algebraic Geometry
40. *Splines on Tetrahedral Decompositions* 05/2016
15th International Conference on Approximation Theory, San Antonio, TX
Minisymposium on Approximation Theory and Algebraic Geometry
41. *Generalized Splines and Graphic Multi-Arrangements* 10/2015
AMS Sectional Meeting, Chicago, IL
Special Session on Combinatorial and Computational Algebra
42. *Piecewise Polynomials and Regularity* 04/2015
Mathematisches Forschungsinstitut Oberwolfach, Germany
Workshop on Multivariate Splines and Algebraic Geometry
43. *Castelnuovo-Mumford Regularity of Mixed Spline Spaces* 01/2015
AMS-MAA Joint Mathematics Meetings, San Antonio, TX
Session on Commutative Algebra
44. *Regularity of Planar Splines* 04/2014
AMS Sectional Meeting, Lubbock, TX
Special Session on Commutative Algebra and Algebraic Geometry
45. *Regularity and Piecewise Polynomial Functions* 04/2014
KUMUNU jr, Lincoln, NE
46. *Local Properties of Splines* 03/2014
Southwest Local Algebra Meeting, College Station, TX
Graduate Student Poster Session
47. *Lattice-Supported Splines on Polytopal Complexes* 01/2014
AMS-MAA Joint Mathematics Meetings, Baltimore, MD
AMS Special Session on Hyperplane Arrangements and Applications
48. *Lattice-Supported Bases for Polyhedral Splines* 08/2013
SIAM Conference on Applied Algebraic Geometry, Fort Collins, CO
Session on Approximation Theory, Geometric Modeling, and Algebraic Geometry
49. *Bivariate Continuous Splines on Polyhedral Complexes* 04/2013
14th International Conference on Approximation Theory, San Antonio, TX
Minisymposium on Multivariate Splines
50. *Shellability and Freeness of Continuous Splines* 10/2012
AMS Sectional Meeting, Tulane, LA
Special Session on Approximation Theory, Geometric Modelling, and Algebraic Geometry
51. *Exploring Gassmann Triples* 01/2009
AMS-MAA Joint Mathematics Meetings
Undergraduate Student Poster Session (\$100 prize)

SEMINAR &
COLLOQUIUM
TALKS

1. *Splines and commutative algebra* 02/2026
Series of two talks for the Algebra Seminar, NMSU
2. *Hyperplane arrangements: combinatorics, the module of derivations and the Jacobian ideal* 02/2025
Series of two talks for the Algebra Seminar, NMSU
3. *A bridge between the algebra and geometry of hyperplane arrangements* 10/2024
Algebra and Geometry Seminar, University of New Mexico, Albuquerque, NM
4. *Inverse systems, symbolic powers, and sequence duality* 09-10/2024
Series of two talks for the Algebra Seminar, NMSU
5. *Connecting the algebra and geometry of line arrangements via rigidity theory* 03/2024
Colloquium, University of Idaho, Moscow, ID
6. *Extremal syzygies of line arrangements and rigidity of planar frameworks* 02/2024
Virtual talk for TATERS seminar hosted by Boise State University, Boise, ID
7. *An introduction to symbolic powers and asymptotics of the containment problem* 02/2024
Series of two talks for the Algebra Seminar, NMSU
8. *Koszul Rees Algebras of Borel Ideals* 08/2023
Algebra Seminar, NMSU
9. *Two perspectives on affine semigroups* 08/2023
Colloquium, New Mexico State University, Las Cruces, NM
10. *Saturating the Jacobian ideal of a line arrangement via rigidity theory* 03/2023
Algebra Seminar, Georgia Institute of Technology, Atlanta, GA
11. *Exploring affine semigroups* 04/2022
Colloquium, University of Texas at Tyler, Tyler, TX (virtual due to COVID-19)
12. *A duality for sequences and its manifestation for symbolic powers* 03/2022
Algebraic Geometry and Geometric Topology Seminar, Tulane University, New Orleans, LA
13. *Homogeneous trivariate splines on vertex stars* 05/2021
Online workshop *Dimension of Multivariate Splines*, University of Rome “Tor Vergata”
14. *Wilf’s conjecture and its extensions* 11/2020
Graduate Seminar, Towson University, Towson, MD (virtual due to COVID-19)
15. *Resurgence via Asymptotic Resurgence* 08/2020
Algebra and Geometry Seminar, Iowa State University, Ames, IA (virtual due to COVID-19)
16. *Extending Wilf’s Conjecture* 10/2019
Colloquium, University of North Carolina-Charlotte, Charlotte, NC
17. *Multi-derivations of hyperplane arrangements* 06/2019
Mediterranea University of Reggio Calabria, Italy
18. *Combinatorics, topology, and algebra of hyperplane arrangements* 06/2019
University of Messina, Italy
19. *Commutative Algebra and Piecewise Polynomials* 02/2018
Colloquium, Marquette University, Milwaukee, WI
20. *Commutative Algebra and Approximation Theory* 01/2018
Colloquium, University of Nebraska-Lincoln, Lincoln, NE
21. *Homological Obstructions to Freeness of Multi-Arrangements* 10/2017
Geometry Seminar, Texas A&M University, College Station, TX
22. *Free Multi-Braid Arrangements and Resolutions* 03/2017
Algebra Seminar, University of Arkansas, Fayetteville, AK
23. *Dimensions of Spline Spaces and Commutative Algebra* 11/2016
Colloquium, Towson University, Towson, MD
24. *Two Tales of Freeness* 11/2016
Colloquium, US Naval Academy, Annapolis, MD
25. *Multi-Derivations of Braid Arrangements* 09/2016
Combinatorics Seminar, University of Kansas, Lawrence, KS
26. *Piecewise Polynomials and Algebraic Geometry* 04/2016
Colloquium, University of Idaho, Moscow, ID

	27. <i>Semialgebraic Splines</i>	03/2016
	Valley Geometry Seminar, University of Massachusetts, Amherst, MA	
	28. <i>Commutative Algebra meets Approximation Theory</i>	11/2015
	Numerical Analysis Seminar, Oklahoma State University, Stillwater, OK	
	29. <i>Commutative Algebra and Approximation Theory</i>	09/2015
	Colloquium, Oklahoma State University, Stillwater, OK	
	30. <i>Splines, Syzygies, and Freeness</i>	09/2015
	Algebra Seminar, Oklahoma State University, Stillwater, OK	
	31. <i>Regularity of Planar Splines</i>	09/2015
	Geometry Seminar, Texas A&M University, College Station, TX	
	32. <i>Algebraic Geometry and Approximation Theory</i>	02/2015
	Colloquium, University of South Florida, Tampa, FL	
	33. <i>Associated Primes of Complexes Arising in Approximation Theory</i>	11/2014
	Commutative Algebra Seminar, UIUC	
	34. <i>Castelnuovo-Mumford Regularity in Approximation Theory</i>	11/2014
	Algebraic Geometry Seminar, UIUC	
	35. <i>Lehmer's Picturesque Exponential Sums with a Twist (with Daniel Schultz)</i>	02/2010
	Number Theory Seminar, UIUC	
TALKS FOR UNDERGRADUATE OR HIGH SCHOOL AUDIENCES	1. <i>Polytopes, graphs, and rigidity</i>	11/2025
	Public Lecture at High School Math Day, Colorado State University, Fort Collins, CO (audience of about 400 high school students and educators)	
	2. <i>Cutting up a pizza and related topics</i>	10/2021
	Colloquium, University of South Alabama, Mobile, AL	
	3. <i>Piecewise Linear Functions, Projecting Polytopes, and Equilibrium Stresses</i>	11/2018
	Symposium of Physics and Mathematics FCFM-IFM, Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Michoacán, Mexico	
	4. <i>Explorations in Rigidity</i>	04/2018
	OSU Math Club, Oklahoma State University, Stillwater OK	
	5. <i>The Best Way to Divide up a Cheese</i>	10/2017
	High School Math Day, Oklahoma State University, Stillwater OK	
	6. <i>The Pizza Cutting Problem</i>	02/2017
	Stillwater High School Math Seminar, Stillwater High School, Stillwater, OK	
	7. <i>Counting Piecewise Linear Functions</i>	03/2016
	Center for Women in Mathematics, Smith College, Northampton, MA	
	8. <i>Jumping Dimensions and Projecting Polytopes</i>	12/2014
	Colloquium, Bradley University, Peoria, IL	
	9. <i>Continuous Piecewise Polynomials and Static Equilibrium</i>	10/2014
	Rose-Hulman Mathematics Seminar, Terra-Haute, IN	
PROFESSIONAL DEVELOPMENT	Participant, NMSU-MÁS (Mejorando las Aulas en STEM / Improving STEM Classrooms) Worked in small teams and attended workshops to learn about and implement active learning techniques in the classroom during fall 2024 and spring 2025. As part of this program, implemented inquiry-based learning (IBL) approach for MATH 1531 - Introduction to Higher Mathematics, in spring 2025. Participant in NMSU PI Academy In fall 2023 and spring 2024, attended workshops on grant-writing and agencies to target for funding opportunities.	
PROFESSIONAL SERVICE	Co-organizer (with Louiza Fouli and Jonathan Montañaño) Arizona-New Mexico Symposium on Commutative Algebra and its Interactions: Geometric Combinatorics, Las Cruces, NM, December 2025	

Co-organizer (with Louiza Fouli and Arvind Kumar)
AMS Special Session on Commutative Algebra and its connections to combinatorics, San Antonio, TX, September 2025.

Co-organizer (with Louiza Fouli and Arvind Kumar)
AMS Special Session on Recent Developments in Commutative Algebra, San Francisco, CA, May 2024.

Organizer
Virtual informal seminar on topics related to splines, Fall 2020-

Co-organizer (with Hendrik Speleers and Deepesh Toshniwal)
Minisymposium on Multivariate Splines: Theory and applications at the International Conference on Approximation Theory and Beyond, Nashville, TN, May 2023.

Co-organizer (with Selvi Kara)
AMS Special Session on Current Trends in Combinatorial and Homological Commutative Algebra, Mobile, AL, November 2021.

Organizer
Postdoc Seminar at CSU, Fall 2020, Spring 2021

Co-organizer (with Nelly Villamizar)
Minisymposium on Algebraic Methods for Multivariate Splines and Rigidity at the SIAM conference on Applied Algebraic Geometry in College Station, Texas, August 2021. (**Virtual due to COVID-19**)

Co-organizer (with Nelly Villamizar)
Minisymposium on Multivariate Spline Approximation and Algebraic Geometry at the SIAM conference on Applied Algebraic Geometry in Bern, Switzerland, July 2019.

Co-organizer (with Frank Sottile)
Minisymposium on Multivariate Splines and Algebraic Geometry at the SIAM conference on Applied Algebraic Geometry in Atlanta, GA, August 2017.

Co-organizer (with Tatyana Sorokina)
Minisymposium on Approximation Theory and Algebraic Geometry at the 15th International Conference on Approximation Theory in San Antonio, TX, May 2016.

Organizer
reading seminar on *The Geometry of Syzygies* in Fall 2011, Spring 2012

Referee

I have served as a referee for articles submitted to the following journals: *Mathematische Annalen*, *Journal of Pure and Applied Algebra*, *International Journal of Algebra and Computation*, *Pacific Journal of Mathematics*, *Constructive Approximation*, *Computer-Aided Geometric Design*, *Journal of Algebraic Combinatorics*, *Graphs and Combinatorics*, *Proceedings of 15th International Conference on Approximation Theory*, *SIGMA*, *Journal of Computational and Applied Mathematics*, *Canadian Mathematical Bulletin*, *Communications in Algebra*, *Épjournal de Géométrie Algébrique*, *Advances in Applied Mathematics*, *Innovations in Incidence Geometry*, *Discrete and Computational Geometry*, *Arkiv för Matematik*, *Collectanea Mathematica*, *Hokkaido Mathematical Journal*, *Journal of Algebra and its Applications*, *ISSAC*, *Mathematics of Computation*, *Selecta Mathematica*, *Journal of Approximation Theory*, *Journal of Algebra*, *Electronic Journal of Combinatorics*.

OTHER AWARDS	Bourgain Fellowship , UIUC	Spring 2013
	REGS Summer Fellowships , UIUC	Summer 2009-2013
	REU Summer Fellowships , UNC Asheville & LSU	Summer 2008-2009
TRAVEL AND CONFERENCE GRANTS	Fields Opportunity for Collaborations US (FOCUS)	02/2026
	collaboration grant from Fields Institute to visit Adam Van Tuyl at McMaster University	
	US Junior Oberwolfach Fellows grant	01/2020
	to attend MFO workshop in Oberwolfach, Germany (not used since the conference was virtual)	

	SIAM Early Career Travel Award	07/2019
	to attend SIAM Conference on Applied Algebraic Geometry in Bern, Switzerland	
	Supported Participant	05/2017
	at CMO Workshop on Symbolic and Ordinary Powers in Oaxaca, Mexico	
	Oberwolfach Leibniz Graduate Students grant	04/2015
	to present at MFO workshop in Oberwolfach, Germany	
	AMS Student Travel Grant	04/2014
	for presentation at AMS Sectional Meeting at Texas Tech	
	AMS Student Travel Grant	01/2014
	for presentation at AMS-MAA Joint Mathematics Meetings	
	Student Travel Award	08/2013
	to attend SIAM Conference on Applied Algebraic Geometry in Fort Collins, CO	
	Travel Award	04/2013
	for presentation at 14th International Conference on Approximation Theory	
	Supported Participant	12/2012
	at MSRI Workshop on Combinatorial Commutative Algebra	
	AMS Student Travel Grant	10/2012
	for presentation at the AMS Sectional Meeting at Tulane	
	Supported Participant	06-07/2012
	at IMA summer school in Applied Algebraic Geometry at Georgia Tech	
SELECTED WORKSHOPS ATTENDED	Fields Institute: Workshop on Applications of Commutative Algebra	05/2025
	Toronto, Canada	
	IPAM: Workshop on Computational Interactions between Algebra,	02/2025
	Combinatorics, and Discrete Geometry	
	Los Angeles, CA	
	INDAM Meeting: Approximation Theory and Numerical Analysis	09/2022
	meet Algebra, Geometry, Topology	
	Cortona, Italy	
	MFO workshop on Logarithmic Vector Fields and Freeness of Divisors	01/2021
	and Arrangements: New perspectives and applications	
	Oberwolfach, Germany	
	Macaulay 2 workshop on coding in the computer algebra system Macaulay2	07/2017
	Berkeley, CA	
	CMO workshop on Ordinary and Symbolic Powers of Ideals	05/2017
	Oaxaca, Mexico	
	Macaulay2 workshop on coding in the computer algebra system Macaulay2	05/2015
	Boise, ID	
	MFO workshop on Multivariate Splines and Algebraic Geometry	04/2015
	Oberwolfach, Germany	
	MSRI workshop on Combinatorial Commutative Algebra	12/2012
	San Francisco, CA	
	IMA summer school in Applied Algebraic Geometry at Georgia Tech	06-07/2012
	Atlanta, GA	
PROFESSIONAL MEMBERSHIPS	American Mathematical Society	